

model. You may take help of Google to search for the kit or parts required to assemble your printer and further instructions, if required. The LMSUU Prusa Mendel 3D printer assembled at EFY required parts that are shown in the box on previous page.

Print a part

If you have done the mechanical assembly and software upload, the major task is over. Now you just have to use Pronterface software to print any 3D object. Start by connecting your 3D printer via USB cable to a computer that has Pronterface software installed already. Connect the ATX power supply to the 3D printer and switch it on.

Run Pronterface software. Once the software is up, computer screen will look like as shown in Fig. 15. Select the COM port to which your 3D printer is connected and set the baud rate to 250,000 (as shown in Fig. 15). Next, click on Connect button. You will see message in the right column of Pronterface indicating that the printer has successfully connected as shown in Fig. 16.

Once connected, you can use Pronterface to manually control the X, Y and Z axis movements as shown in Fig. 17. You can also manually control the extruder motor using Pronterface, but be sure that the heater is turned on before doing so. The distance and speed of the manual control is set right below the Extrude button.

To set the temperature of the nozzle or heat-bed, use the pull-down menu and select the temperature and then use Set button as shown in Fig. 18. To turn the heating off, simply select the Off button. You can monitor the temperature of

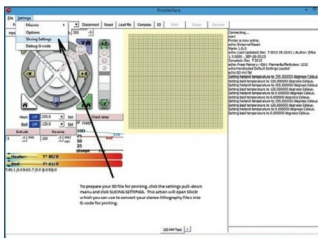


Fig. 19: Selection of Slicing Settings



Fig. 20: Slice3r configuration wizard

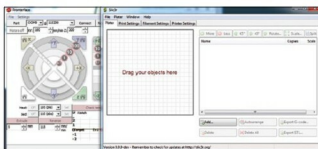


Fig. 21: Dragging the .stl file

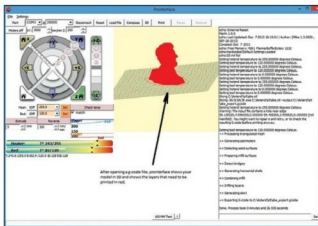


Fig. 22: Loading G-code file

the nozzle and bed by selecting 'Watch' option and then viewing the graph of real-time temperature value. Or, you can select 'check temp' to view the actual temperature in the console on the right side.

From the pull-down menu in the toolbar select Settings and then Slicing Settings as shown in Fig. 19. This will open the program Slic3r as shown in Fig. 20. Slic3r is the program to convert a digital 3D model into printing instructions for your 3D printer.

If this is the first time you are opening Slic3r, you will be prompted with the Slic3r configuration wizard. Select Cancel on this wizard.

Drag your .stl file of the object you want to print in the box as shown in Fig. 21. Click on the export G-code button and your G-code will be ready to load. Your 3D printer uses G-code to know what to print. The slicing program will convert a stereolithography (.stl) file of a 3D model into the necessary G-code that your printer can understand. Once the file has been exported into the needed G-code format, you can open that file in Pronterface for printing.

Open the G-code by navigating File menu and then clicking Open. Browse the folder you exported the G-code file into and select that file and then press OK. When Pronterface loads the file it will appear on the graph in the graphical user interface as shown in Fig. 22. Now you just have to click on Print on the Pronterface toolbar and your printer will start printing the 3D object, layer by layer.

Happy printing! ●

The author was a technical editor at EFY when he wrote this article. He has moved on since then to work on his own